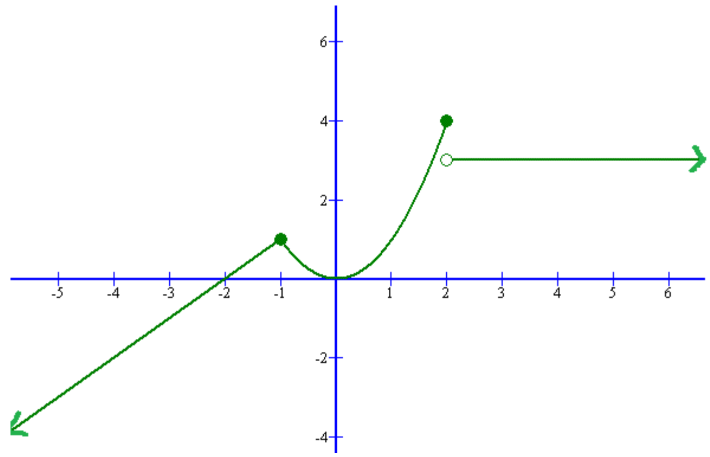
MTH 200 Chapter 2: Limits

2.3.1 Limits Worksheet

1. Find the following limits (if they exists). If the limit does not exist explain why.

   a. Find $\lim\limits_{t \to -1} f(t)$

   b. Find $\lim\limits_{t \to 2^+} f(t)$

   c. Find $\lim\limits_{t \to 2} f(t)$

Problems #2-12: Evaluate the limit, if it exists. If you claim that is does not exist (DNE) then provide a valid argument to support your claim.

2. $\lim\limits_{x \to -4} \dfrac{x^2 + 5x + 4}{x^2 + 3x - 4}$

3. $\lim\limits_{x \to 4} \dfrac{x^2 - 4x}{x^2 - 3x - 4}$

4. $\lim\limits_{x \to -1} \dfrac{x^2 - 4x}{x^2 - 3x - 4}$

5. $\lim_{h \to 0} \frac{\sqrt{1+h}-1}{h}$

6. $\lim_{t \to 0} \left(\frac{1}{t} - \frac{1}{t^2+t} \right)$

7. $\lim_{x \to 2.5^+} \frac{e^x}{5-2x}$

8. $\lim_{x \to 16} \frac{4-\sqrt{x}}{16x-x^2}$

9. $\lim_{x \to -1} \frac{x^2+2x+1}{x^4-1}$

10. $\lim_{x \to -2} \frac{2-|x|}{2+x}$ (*Hint: If* $x \to -2$ *what can you say about the value of* $|x|$ *?*)

11. Consider the function $f(x) = \begin{cases} 1+\sin(x) & if \;\; x<0 \\ \cos(x) & if \;\; 0 \le x \le \pi \\ \sin(x) & if \;\; x > \pi \end{cases}$.

a) Find $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

b) Does $\lim_{x\to 0} f(x)$ exist?

c) Find $\lim_{x\to \pi^-} f(x)$ and $\lim_{x\to \pi^+} f(x)$.

d) Does $\lim_{x\to \pi} f(x)$ exist?

12. Consider the function $g(x) = \begin{cases} x^2 & if \;\; x<1 \\ \sqrt{x} & if \;\; x \ge 1 \end{cases}$.

a) Find $\lim_{x\to 0} g(x)$.

b) Find $\lim_{x\to 1} g(x)$.

13. Use the Squeeze Theorem to help determine the following limit: $\lim_{x\to 0} x^3 \cos\left(\frac{1}{x^2}\right)$.

14. Use the Squeeze Theorem to help determine the following limit: $\lim_{x\to -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10}$.